

Diagnostic Evaluation Report

1. Framework Information

- **Framework:** X-RAG Diagnostic Framework (v1.0)
- **Trace ID:** `b5ca7713-d0e2-4238-8f1b-3bffffdf29c5f`
- **Timestamp:** 2026-09-05T16:11:34.955352Z
- **Status:** COMPLETED

2. Executive Summary

- **Question:** Which judgments in this corpus apply or construe Section 178?
- **Answer:** The retrieved corpus contains the text of **Section 178** (see [Chunk-ID: 8fca1873-e179-4afb-a429-c88047cb5cba]), but it does not include any judicial judgments that apply or construe that section. 178 is present in the provided chunks.

Therefore, I do not have enough information to identify any judgments that apply or construe Section 178.

- **Health Score:** 0.56
- **Primary Issue:** GROUNDING_FAILURE
- **Claims Supported:** 1 / 4
- **Summary:** Analysis completed.

3. Pipeline Overview

Stage	Status	Confidence	Observation
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CORPUS	PASS	0.9	Corpus contains sufficient relevant content, evidenced by at least one supported claim.
RETRIEVER	PASS	0.95	At least one retrieved chunk provides supporting evidence for a verified claim.
GENERATOR	PASS	0.95	All generated claims are fully supported by evidence.
GROUNDING	FAIL	0.95	2 claim(s) are actively contradicted by retrieved evidence.

Efficiency: 1/6 chunks used (17%)

4. Evaluation Metrics

- **Retrieved Chunks:** 6
- **Verified Claims:** 4 (Supported: 1, Partial: 0, Unsupported: 0, Contradicted: 2)
- **Grounding Score:** 0.25
- **Evidence Coverage:** 0.50
- **Average Entailment:** 0.30
- **Latencies:** Retrieval (2291.3ms) | Generation (13266.6ms) | Verification (16487.0ms)

5. RAGAS Metrics

- **Faithfulness:** 0.2500
- **Answer Relevancy:** 0.8850
- **Context Precision:** 1.0000
- **Context Relevancy:** 0.0000
- **Context Recall:** N/A
- **Answer Similarity:** N/A
- **Answer Correctness:** N/A

Answer Correctness (Claim Recall)

Not computed for this trace.

6. Evidence Analysis

- **[44aea539-ead8-4b98-8de2-d5036ab4fd36]** (SUPPORTED): The retrieved corpus contains the text of Section 178 (see [Chunk-ID: 3073ca28-3dae-4ea9-ac2d-61f2230c4474]): *He produces from his custody deeds relating to the land showing his titles to it.
- **[a746edea-ad64-4dfb-99c2-727640b63b04]** (CONTRADICTED): The retrieved corpus does not include any judicial judgments that apply or construe that section. *Presumption as to due execution, etc., of documents not produced.
- **[67b70ffc-1676-41a7-9e9c-0ca30ea446fa]** (NOT_VERIFIABLE): No case law or judicial interpretation of Section 178 is present in the provided corpus. *2 of 2016.

- **[5347172a-57a6-4833-8b76-86f4b5a1a0bd]** (CONTRADICTED): I do not have enough information to identify any judgments that are supported by the evidence.
- Evidence (Chunk 4a051196-d494-4463-9e89-daf8fa2ea57c): 32.

7. Root Cause Analysis

- **Primary Cause:** GROUNDING_FAILURE
- **Confidence:** 0.94
- **Reasoning:**
 - Starting root cause analysis for trace b5ca7713-d0e2-4238-8f1b-3bffffdf29c5f.
 - CORPUS passed (confidence=0.90).
 - RETRIEVER passed (confidence=0.95).
 - Skipping CHUNKING: no observable state in matrix.
 - GENERATOR passed (confidence=0.95).
 - Failure at GROUNDING: GROUNDING_FAILURE (confidence=0.95). Observation: 2 claim(s) are actively contradicted by retrieved evidence.
 - Primary root cause: GROUNDING_FAILURE at GROUNDING (earliest upstream failure, confidence=0.95).
 - Analysis complete. Primary cause: GROUNDING_FAILURE.

8. Corrective Action Plan

- **[EXPERIMENTAL] Require Explicit Citations**
 - Description: Force the LLM to append [Chunk-ID] to every sentence it generates.
 - Expected Improvement: Self-correction during generation ensures every fact is linked to evidence. (Metric: 100% of claims supported)
- **[INFORMATIONAL] Review Retrieval Top-K / Reranker Cutoff**
 - Description: Retrieval efficiency is low (17%); consider reducing top-k or tightening the reranker score cutoff.
 - Expected Improvement: Reduced prompt size and lower latency/cost with no loss of verified support. (Metric: Chunk utilization)

9. Evidence Traceability

- **Primary Cause:** GROUNDING_FAILURE
- **Derived From:** RootCauseAnalysis
- **Pipeline Stage:** GROUNDING
- **Supporting Claims:** 44aea539-ea8d-4b98-8de2-d5036ab4fd36, a746edea-ad64-4dfb-99c2-727640b63b04, 67b70ffc-1676-41a7-9e9d-3073ca28-3dae-4ea9-ac2d-61f2230c4474
- **Supporting Chunks:** 3073ca28-3dae-4ea9-ac2d-61f2230c4474, 3073ca28-3dae-4ea9-ac2d-61f2230c4474, 6888741c-cf31-4107-b4d0-3073ca28-3dae-4ea9-ac2d-61f2230c4474

10. Overall Assessment

- **Major Strength:** RETRIEVER is healthy (confidence=95%): At least one retrieved chunk provides supporting evidence for the claim.
- **Major Weakness:** GROUNDING_FAILURE
- **Next Priority:** Require Explicit Citations
- **Recommendation:** Pipeline requires targeted improvements to the primary failure stage.

11. Framework Footer

- **Generated By:** X-RAG Diagnostic Framework
- **Artifact Version:** 1.0

APPENDIX

Artifacts Generated

✓ RAGTrace

artifacts/rag_traces/b5ca7713-d0e2-4238-8f1b-3bffffdf29c5f.json

✓ ClaimSet

artifacts/claim_sets/b5ca7713-d0e2-4238-8f1b-3bffffdf29c5f.json

✓ VerificationResults

artifacts/verification/b5ca7713-d0e2-4238-8f1b-3bffffdf29c5f.json

✓ PipelineStateMatrix

artifacts/pipeline_state_matrix/b5ca7713-d0e2-4238-8f1b-3bffffdf29c5f.json

✓ RootCauseAnalysis

artifacts/root_cause_analysis/b5ca7713-d0e2-4238-8f1b-3bffffdf29c5f.json

✓ CorrectiveActionPlan

artifacts/corrective_action_plan/b5ca7713-d0e2-4238-8f1b-3bffffdf29c5f.json

✓ DiagnosticEvaluationReport

artifacts/reports/b5ca7713-d0e2-4238-8f1b-3bffffdf29c5f.json